

Project Context: HimShakti Bio-Waste & Processing Analytics

Part 1:

AI-Generated Insight Prompts & Critical Evaluation (Target: 5 Prompts)

1. **AI Question 1:** *"Which processing center has the highest total volume of crop waste mass in KG?"*
 - **Evaluation:** **Accurate**
 - **Reason:** Fully answerable because our dataset tracks Waste_Mass_KG across individual Processing_Center variables.
2. **AI Question 2:** *"What is the exact correlation between the fulfillment cycle time in hours and the financial loss in INR?"*
 - **Evaluation:** **Partially Accurate**
 - **Reason:** Both columns exist, but a simple correlation misses external variables like changing market raw crop prices.
3. **AI Question 3:** *"How much total financial loss in INR is specifically caused by Herbal Extract compared to Ginger and Turmeric?"*
 - **Evaluation:** **Accurate**
 - **Reason:** Directly verifiable; our dashboard donut chart tracks loss percentages by material type.
4. **AI Question 4:** *"Which specific agricultural farm supplier provided the batch that caused the highest processing waste spike?"*
 - **Evaluation:** **Inaccurate**
 - **Reason:** The operational dataset tracks regional centers, but completely lacks a data column for individual farm suppliers.
5. **AI Question 5:** *"What is the monthly percentage growth trend of biomass yields across all centers over the last two years?"*
 - **Evaluation:** **Partially Accurate**
 - **Reason:** We can track monthly trends using the Date column, but our dataset only covers the current Q2 window, not two full years.

Part 2:

AI Narrative Fact-Checking & Edit Log (Target: 2-3 Examples)

Example 1: Product Loss Contribution

- **Before (AI Claim):** *"Herbal Extract is responsible for nearly all financial leakages across HimShakti processing centers, completely eclipsing other crops due to a 90% waste rate."*
- **After (Edited Narrative):** *"Herbal Extract is the primary loss driver, accounting for 56.4% of total financial leakages, representing a major portion of the 17,203,520 INR aggregate loss."*
- **Reason for Edit:** The AI hallucinated a "90% waste rate" and overstretched the impact. The actual dashboard data shows it accounts for 56.4% of the total financial value lost.

Example 2: Processing Center Performance

- **Before (AI Claim):** *"Center-A is failing across every operational metric, doubling the fulfillment cycle times of all neighboring facilities due to equipment failure."*
- **After (Edited Narrative):** *"Center-A handles the highest volume of inbound mass, resulting in higher absolute waste mass, though its processing cycle times remain comparable to regional averages."*
- **Reason for Edit:** The AI assumed higher absolute waste meant complete equipment failure and double the cycle times. Data verification proved Center-A simply processes a much larger baseline volume of raw material.